TGE

Release 1.0.0

AE,SB,BB

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Contents:

CHAPTER

ONE

GRID MODULE

1.1 Tapered Gridded Estimator (TGE)

For a given **input uv-fits file**, we taper the sky response of the telescope by convolving visibilities with a function $\tilde{w}(\vec{\mathbf{U}}) = \pi\theta_w^2 \mathbf{exp}[-\pi^2\theta_w^2 \mathbf{U}^2]$ (which is the Fourier Transform of a window function $\mathcal{W}(\vec{\boldsymbol{\theta}}) = \mathbf{exp}[-\theta^2/\theta_w^2]$ which falls off to a value close to zero well before the first null of the telescopes primary beam pattern) and gridded on a square grid in the uv-plane using,

$$\boxed{\mathcal{V}_{cg}^P(\vec{\mathbf{U}}_g, \nu_a) = \sum_i \tilde{w}(\vec{\mathbf{U}}_g - \vec{\mathbf{U}}_i)\ \mathcal{V}^P(\vec{\mathbf{U}}_i, \nu_a)\ F_i^P(\nu_a)} \tag{1.1}$$

The sum is over all the baselines. Here ν_a refers to the different frequency channels, $\vec{\mathbf{U}}_i$ is i^{th} baseline. $\mathcal{V}_{cg}^P$ refers to the convolved visibility at grid point g and by the superscript P we denote the polarization. $\vec{\mathbf{U}}_g$ is the baseline corresponding to this grid point and $F_i^P(\nu_a)$ incorporates the flagging information of the data corresponding to the frequency ν_a and for the corresponding polarization P. P can take be either in $\mathrm{XX}, \mathrm{YY}, \mathrm{XY}, \mathrm{YX}$ basis or in the circular basis $\mathrm{RR}, \mathrm{LL}, \mathrm{RL}, \mathrm{LR}$, based on the telescope.

$F_i^P(\nu_a)$ has a value 0 if the data at a given baseline and frequency is flagged and 1 otherwise. Note that the baselines $\vec{\mathbf{U}}_i$ here are defined at a fixed reference frequency ν_c (**centered frequency of the fits file**), and in this analysis they are considered to be fixed as we vary the frequency channel ν_a (although in principle they are not).

In the visibility data, we flag the data if the corresponding weight value is less than 0, otherwise we take it as it is.

We calculate convolved visibility $\mathcal{V}_{cg}^P(\vec{\mathbf{U}}_g, \nu_a)$ for a given no of frequency channels and for given polarizations of the data. A 4D array is returned, where **first one is the polarization axis**, next **two dimensions are values of convolved visibility at each grid points** (square grid) in uv plane and **last one is frequency axis**.

Here $\theta_w = f\theta_0$, with $\theta_0 = 0.6\,\theta_{\mathrm{FWHM}}$. f is known as tapering parameter whose value can be suitably chosen. The effect of tapering is enhanced if the value of f is reduced. For instance $f < 1$ would suppress the sky response from the outer regions and the side-lobes of the PB pattern, whereas a large value $f > 1$ would imply very little tapering. Preferably $f \leq 1$ so that $\mathcal{W}(\vec{\boldsymbol{\theta}})$ cuts off the sky response well before the first null of the primary beam.

The grid spacing in the uv plane is given by $\Delta \mathrm{U} = \dfrac{\sqrt{ln2}}{2\pi\theta_{\mathrm{eff}}}$ with $\theta_{\mathrm{eff}} = \dfrac{f\theta_0}{\sqrt{1+f^2}}$.

Meanwhile, those baselines are rejected for which $|\vec{\mathbf{U}}_i| < |\vec{\mathbf{U}}|_{max}$. We effectively increase the size of visibility data twice ; original one and one for all the baselines with $-\vec{\mathbf{U}}_i$ as $\mathcal{V}^P(-\vec{\mathbf{U}}_i, \nu_a) = \left[\mathcal{V}^P(\vec{\mathbf{U}}_i, \nu_a)\right]^*$.

For any baseline $\vec{\mathbf{U}}_i$, the contribution of visibilities to the grid points is taken for only those grid points for which are within(and on) a square grid of side length $12\Delta\mathrm{U}$ whose center is at grid point which is nearest to $\vec{\mathbf{U}}_i$ (The sides of this grid are along u and v axis). The weight function $\tilde{w}(\vec{\mathbf{U}}_g - \vec{\mathbf{U}}_i)$ falls considerably faster and we do not expect a significant contribution from the visibilities beyond this baseline separation.

Please note that in this analysis Baseline migration is not incorporated.

1.2 Module Functions

`grid.mkbin`(*GV*, *dU*, *Umax*, *FWHM*, *f*, *binUmax*, *binUmin*, *Nbin*, *Mg_min*, *outf*)

Given a **UAPS convolved gridded visibility array** and suitable gridding parameters it identfies the grid points which are relevant and which grid points are in which bin. Also the effective multipole value for a particular bin is given by :

$$\ell_a = \frac{\sum_g w_g \, \ell_g}{\sum_g w_g}; \quad \text{with}\, w_g = M_g$$

And $\ell_g = 2\pi|\vec{\mathbf{U}}|_g$

Parameters

- **GV** (*np.ndarray*) – Convolved gridded visbility array for the **UAPS** visibilities. Different realizations have been put along frequency axis.
- **dU** (*float*) – Grid spacing ΔU in wavelength units.
- **Umax** (*int or float*) – The maximum value of baseline $|\vec{\mathbf{U}}|_{max}$. Baselines greater than this are rejected.
- **FWHM** (*int or float*) – The full width half maximum (FWHM) of telescopes primary beam pattern.(in degrees)
- **f** (*int or float*) – Tapering parameter value.
- **binUmax** (*int or float*) – Maximum value of $|\vec{\mathbf{U}}|$ (in units of wavelength λ) to be used in the binning infomation. Grid points having baseline greater than this are rejected.
- **binUmin** (*int or float*) – Minimum value of $|\vec{\mathbf{U}}|$ (in units of wavelength λ) to be used in the binning infomation. Grid points having baseline lesser than this are rejected.
- **Nbin** (*int*) – Number of annular bins the whole uv plane has been divided into.
- **Mg_min** (*int or float*) – Minimum value of normalization factor M_g to be used. Grid points at which visibility self correlation is less than this are rejected.
- **outf** (*string*) – The name of file in which the binning information is saved. This is saved in **.npz** format.

Notes

- dU is the grid spacing in baseline units (λ).
- NI is a 2D matrix, which contains the information which grid points are relevent. It contains value $-2, -1, 0, \cdots, N_{bin} - 1$. Grid points those satisfy $NI \geq 0$ are relevant. The rest of them are rejected in our analysis.
- ix is a 1D array, which contains the u index of grid points in terms of grid spacing dU, i.e. $ix_g = u_g/d$U. (only for relevant grid points).
- iy is a 1D array, which contains the v index of grid points in terms of grid spacing dU, i.e. $iy_g = v_g/d$U. (only for relevant grid points).

- ni is a 1D array, which contains information about which grid point falls into which annular bin. If the value is zero then it is residing in first bin, if 2 then in second bin and so on. Max value is $N_{bin} - 1$. (only for relevant grid points).

Examples

```
>>> # BIN INFO
>>> import grid as gd
Imported Grid, Import Time : 19/05/26 | 07:48:35 PM
>>> # ===================================================
>>> # Set the gridding parameters in the TGE code
>>> n1      = 0     # Starting channel number
>>> n2      = 99    # End channel number
>>> Flag    = False # Apply actual flagging of data
>>> Umax    = 250   # Baselines greater than this are rejected
>>> FWHM    = 23    # Primary beam FWHM of the telescope in degrees
>>> f       = 0.6   # Tapering parameter value
>>> nstokes = [0]   # 0 for XX and 1 for YY (polarizations to grid)
>>> # ===================================================
>>> nrel    = -1    # grid along freq axis
>>> # For binning
>>> Nbin    = 20    # No of annular bins the whole uv plane has been divided into
>>> binUmin = 6.0   # min |U| used in getting binning information
>>> binUmax = 220   # max |U| used in getting binning information
>>> Mg_min  = 0.01  # M_g cutoff, grid points have less than this will be rejected
>>>
>>> inpath   = '/home/baijayanta/Complete_Pipeline/uaps_files/'
>>> filename = '1000007200_out_uaps_15.fits'
>>> # name of input .fits file for which we will calculate the bin info
>>> infile   = inpath + filename
>>> # grid it # ([Nstokes,Nu,Nv,Nc],(dU,Umax,FWHM,f))
>>> GV, info = gd.grid(infile,n1,n2,nrel,Umax,FWHM,f,Flag, nstokes)
<<<<<<<<<<< TGE >>>>>>>>>>>>
<< 19/05/26 | 07:48:35 PM >>
Type  : Data
<< 19/05/26 | 07:48:48 PM >>
Elapsed   :   00:00:12.82
<<<<<<<<<<< Done >>>>>>>>>>>
>>> # make bin info
>>> gd.mkbin(GV[0], info[0], info[1], info[2], info[3], binUmax,
...                                  binUmin, Nbin, Mg_min, outf = 'bin_info_7200')
<<< Bin info >>>
dU   : 1.07
Umax : 250
FWHM : 23 degrees
f    : 0.6
Shape: (457, 457, 100)
data : 22.235
Saved: bin_info_7200.npz
```

`grid.grid`(*infile*, *n1*, *n2*, *nrel*, *Umax*, *FWHM*, *f*, *Flag*, *nstokes*, *seed=None*)

This tapers the sky response and grids the uv plane and computes convolved visibilities, given a fits file.

$$\boxed{\mathcal{V}_{cg}^{P}(\vec{\mathbf{U}}_g, \nu_a) = \sum_i \tilde{w}(\vec{\mathbf{U}}_g - \vec{\mathbf{U}}_i)\ \mathcal{V}^P(\vec{\mathbf{U}}_i, \nu_a)\ F_i^P(\nu_a)} \tag{1.2}$$

Parameters

- **infile** (*Fits object*) – Fits file which contains all the data.
- **n1** (*int*) – Starting frequency channel number.
- **n2** (*int*) – End frequency channel number.
- **nrel** (*int*) –

1. If positive then it copies that frequency channel data to all the frequency channels. Flagging Information is same as original fits file.
2. If -1 then leaves the data as is.
3. If -2 then simulates random gaussian noise visibilities for the baselines with mean $\mu = 0$ and std $\sigma = 1$.

- **Umax** (*int or float*) – The maximum value of baseline $|\vec{\mathbf{U}}|_{max}$. Baselines greater than this are rejected.
- **FWHM** (*int or float*) – The full width half maximum (FWHM) of telescopes primary beam pattern.(in degrees)
- **f** (*int or float*) – Tapering parameter value.
- **Flag** (*boolean type (True or False)*) – If True it flags the visibility data(puts the data to zero), else does nothing.
- **nstokes** (*int or list*) – The int or the list elements must be within the range of polarizations in the original fits file.
- **seed** (*int or None*) – Default set None, seed value for noise only simulations.

Returns

The tuples first element is a 4D convolved gridded visibility array, where first dimension is polarization axis, next two dimensions are values of convolved visibility at each grid points (square grid) in uv plane and last one is along frequency axis. Next element is in the form : (dU, Umax, FWHM, f).

Return type

Tuple

Examples

```
>>> import grid as gd
Imported Grid, Import Time : 19/05/26 | 07:28:16 PM
>>> # Set the gridding parameters in the TGE code
>>> n1        = 0     # Starting channel number
>>> n2        = 767   # End channel number
>>> Umax      = 250   # Baselines greater than this are rejected
>>> FWHM      = 23    # Primary beam FWHM of the telescope in degrees
>>> f         = 0.6   # Tapering parameter value
>>> Flag      = True  # Apply actual flagging of data
>>> nstokes   = [0,1] # 0 for XX and 1 for YY (which polarizations you want to grid)
>>> infile    = '/home/MWA/data_cutoff/1000000000.fits' # fits file name
>>> nrel      = -1  # data file
>>> GV, info = gd.grid(infile, n1, n2, nrel, Umax, FWHM, f, Flag, nstokes)
<<<<<<<<<<< TGE >>>>>>>>>>>>
<< 19/05/26 | 07:28:16 PM >>
Type  : Data
<< 19/05/26 | 07:30:05 PM >>
Elapsed   :   00:01:49.04
<<<<<<<<<<< Done >>>>>>>>>>>
>>> print(f"{GV.shape = }")
GV.shape = (2, 457, 457, 768)
>>> print(f"dU   : {info[0]:.2f}\
...       \nUmax : {info[1]}\
...       \nFWHM : {info[2]} degrees\
...       \nf    : {info[3]}")
dU   : 1.07
Umax : 250
FWHM : 23 degrees
f    : 0.6
```